9A Complementary Events, Venn Diagrams and Two-Way Tables

Standard Worksheet

Name: _____ Class: _____ Date: _____

PART A · FLUENCY & SKILL PRACTICE

Show complete step-by-step working in your workbook

Q1. Use the complementary event rule $\Pr(A') = 1 - \Pr(A)$ to find $\Pr(A')$ for each event:

(a) $\Pr(A) = 0.27$ (b) $\Pr(A) = \dfrac{5}{12}$ (c) $\Pr(A) = 64\%$ (d) $\Pr(A) = \dfrac{19}{25}$

Q2. In a Venn diagram with $\xi = 50$ students, $n(A) = 28$, $n(B) = 22$, and $n(A \cap B) = 12$. Find:

(a) $n(A \text{ only}) = n(A \cap B')$ (b) $n(B \text{ only}) = n(A' \cap B)$ (c) $n(A \cup B)$ (d) $\Pr(A' \cap B')$ (neither A nor B)

Q3. Given $\Pr(A) = 0.60$, $\Pr(B) = 0.50$, and $\Pr(A \cap B) = 0.30$, calculate:

(a) $\Pr(A \cup B)$ (b) $\Pr(A \cap B')$ (A only) (c) $\Pr(A' \cap B)$ (B only) (d) $\Pr(A' \cap B')$ (neither A nor B)

Q4. A letter is chosen at random from the 10 letters of the word AUSTRALIAN. Find the probability of selecting:

(a) The letter A (b) A vowel (A, E, I, O, U) (c) A letter that is NOT A (d) A consonant that is NOT N

PART B · PROBLEM-SOLVING & APPLICATIONS

Write clear mathematical statements with units

Q5. In a group of 60 tourists visiting Phillip Island, 38 visit the Penguin Parade (P), 29 visit the Koala Conservation Reserve (K), and 8 visit neither attraction. Find (a) how many tourists visit BOTH attractions and (b) the probability that a randomly selected tourist visits the Penguin Parade only.

Q6. A 2×2 two-way table records 120 Year 8 students who either walk to school (W) or take public transport (W'), and whether they wear a blazer (B) or jumper (B'): $n(W \cap B) = 28$, $n(W) = 45$, and $n(B) = 72$. Complete the table counts and find $\Pr(W' \cap B')$.

Q7. A spinner with 4 equal sectors numbered $1, 2, 3, 4$ is spun twice (16 equally likely outcomes). Find the probability that (a) the sum of the two spins is 6, and (b) at least one of the two spins is a 4.

Q8. A box of chocolates contains 6 dark chocolates and 4 milk chocolates (10 in total). Two chocolates are eaten one after the other without replacement. Find the probability that both chocolates are dark.

PART C · REASONING & CHALLENGE

Justify your conclusions using algebraic or geometric reasoning

Q9. For two events A and B, $\Pr(A) = 0.45$ and $\Pr(B) = 0.35$. What is the maximum possible value of $\Pr(A \cup B)$ (when A and B are mutually exclusive), and what is the minimum possible value of $\Pr(A \cup B)$ (when B is entirely inside A)?

Q10. Three fair coins are tossed simultaneously (8 equally likely outcomes). Explain why $\Pr(\text{at least one Head and at least one Tail}) = 1 - \Pr(\text{HHH}) - \Pr(\text{TTT}) = \dfrac{3}{4}$.